DATA236 - Lab 1

Yelp Prototype - FastAPI + ReactJS + Agentic AI

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Github repo : https://github.com/Sayani0706/DATA_236_LAB1

Introduction

For this lab, we built a Yelp-style restaurant discovery and review platform using FastAPI for the backend and React for the frontend. The application supports two user types : regular users who can search restaurants, write reviews, save favourites, and interact with an AI chatbot, and restaurant owners who can manage listings through the backend API. The owner frontend pages are optional stretch features and are fully supported through the backend endpoints.

System Design

The app follows a standard client-server architecture. The frontend talks to the backend through REST APIs, and the backend handles all the business logic and database operations.

Layer	Technology
Frontend	React.js, Bootstrap, Axios
Backend	Python, FastAPI
Database	MySQL with SQLAlchemy ORM
Auth	JWT tokens, bcrypt for passwords
AI Assistant	LangChain, Groq LLM (llama-3.3-70b), Tavily Search
API Docs	Swagger UI at /docs (auto-generated by FastAPI)

Backend Structure

The backend is organized like this:

- app/routes/ - all the API route files (auth, users, restaurants, reviews, favorites, owner, ai_assistant)
- app/models/ - SQLAlchemy models that map to MySQL tables
- app/utils/ - JWT token creation and password hashing
- app/config.py - reads environment variables from .env
- main.py - FastAPI app entry point, registers all routers, sets up CORS

Frontend Structure

The frontend is organized into:

- pages/ - full page components (Dashboard, Explore, Login, Signup, Profile, RestaurantDetails, AddRestaurant, UserActivity)
- components/ - reusable components (ChatWindow, RestaurantCard, Navbar, ThinkingIndicator, LoadingSpinner)
- services/ - API layer (api.js with Axios, authService.js, chatService.js, restaurantService.js)
- context/ - AuthContext.js for global token state

All API calls go through a centralized Axios instance in services/api.js which automatically attaches the JWT token to every request and redirects to login on 401 errors.

Database Design

The application uses a relational MySQL database with SQLAlchemy ORM. The schema is designed to support users, restaurants, reviews, personalization, and AI chat history.

Tables

- ➢ **users** – stores user and owner account information
- ➢ **user_preferences** – stores user preferences (cuisine, price, location, etc.)
- ➢ **restaurants** – stores restaurant details and owner information
- ➢ **restaurant_photos** – stores restaurant images
- ➢ **reviews** – stores user reviews and ratings
- ➢ **review_photos** – stores images attached to reviews
- ➢ **favorites** – stores saved restaurants for each user
- ➢ **chat_history** – stores AI conversation messages

Relationships

- ➢ A user can write multiple reviews and save multiple favorites
- ➢ A restaurant can have multiple reviews and photos
- ➢ A review can have multiple photos
- ➢ Each user has one set of preferences
- ➢ Chat history is linked to users for multi-turn AI conversations

Features Implemented

User Features

- Signup and Login - users sign up with name, email and password. Passwords are hashed with bcrypt before storing. Login returns a JWT token.
- Profile - users can view and update name, email, phone, city, state (must be abbreviated like CA), country (dropdown), languages, gender, about me and upload a profile picture.
- Preferences - users can save AI preferences including cuisine, price range, location, search radius, dietary needs, ambiance and sort preference. These are loaded by the AI on every chat.
- Restaurant Search - the Explore page lets users search by name, cuisine, keyword and city. Results can be sorted by rating, popularity or price. Sort triggers automatically without clicking Search again.
- Restaurant Details - each restaurant has its own page showing name, cuisine, address, hours, contact, description, average rating, review count and all reviews with reviewer names and dates.
- Add Restaurant - logged-in users can create new listings with all the required fields.
- Reviews - users can add reviews with 1-5 star rating, comment text and optional photo. Review date is server-generated. Users can only edit or delete their own reviews.
- Favourites - users can save restaurants and view them in the Favourites tab under My Activity.
- History - the My Activity page shows two tabs. History shows all reviews written and restaurants added. Favourites shows saved restaurants.
- AI Assistant - chatbot on the home screen with quick action buttons, multi-turn conversation, thinking indicator, clear chat button and clickable restaurant cards that link to the details page.

Owner Features

- Owner signup with name, email, password and restaurant location
- Post restaurants with full details including pricing tier, amenities, cuisine, contact and hours
- Claim existing restaurant listings to take ownership
- View all reviews for their restaurants in read-only mode
- Owner dashboard showing average rating, review count and recent reviews per restaurant

AI Implementation

The AI assistant uses a 4-step pipeline on every message:

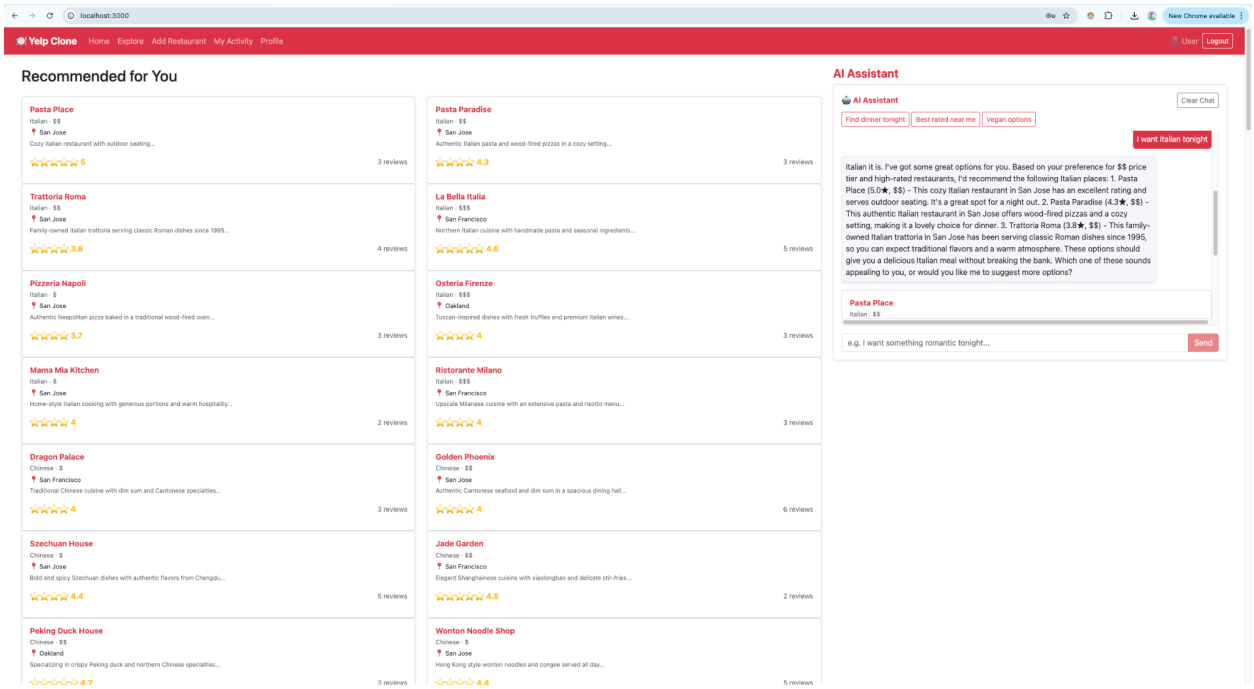
1. Filter Extraction - the user message is sent to Groq LLM through LangChain with a structured prompt that asks it to return a JSON object with cuisine, dietary, ambiance, price, city and keyword. This lets natural language like "I want something vegan and casual" get turned into structured filters.
2. Tavily Web Search - before calling the main LLM, the system runs a Tavily search using the user's query to get current context like trending restaurants or special events. This gets injected into the system prompt so the AI can reference real-world information.
3. Database Query - the extracted filters are used to query the MySQL restaurant database. The user's saved preferences are also loaded and used as fallback filters if nothing was specified in the message.
4. LLM Response - the matching restaurants, user preferences and Tavily web context are all injected into the system prompt. The LLM generates a conversational response with reasoning for each recommendation. The full conversation history is passed on every request to support multi-turn conversations.

The endpoint is POST /ai-assistant/chat. Input is { "message": "...", "conversation_history": [...] }. Output is { "response": "...", "recommendations": [...], "web_context": "..." }. The recommendations array has structured restaurant data (id, name, cuisine, city, pricing_tier, avg_rating) which the frontend uses to show clickable cards.

Results

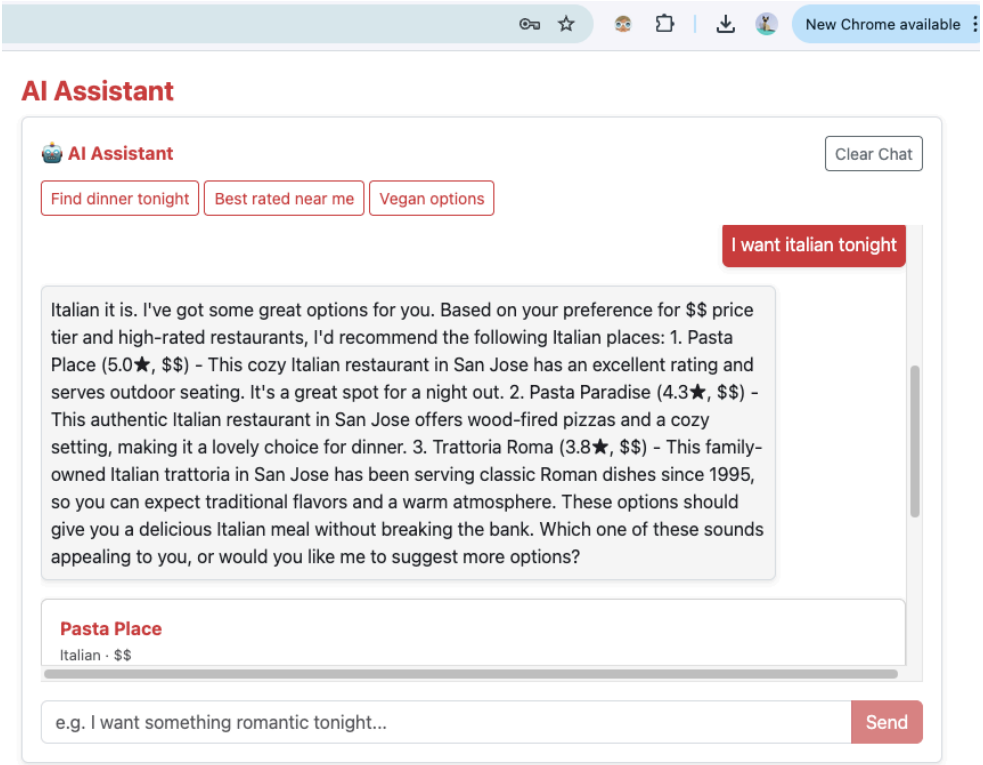
Home Page with AI Assistant

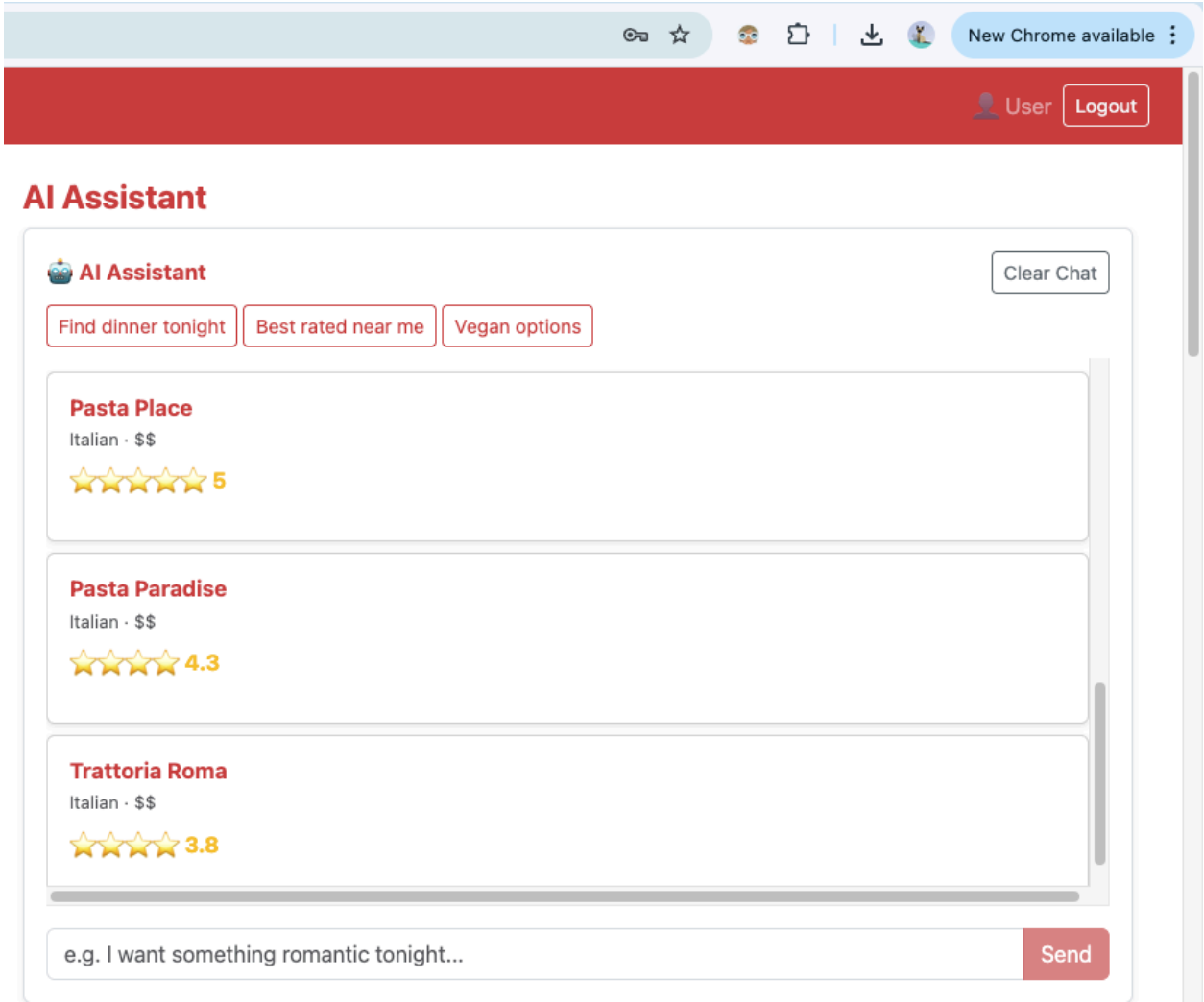
The home page shows restaurant cards on the left and the AI assistant panel on the right. It is always visible and has quick action buttons.



AI Chat - Single Turn

Typed a query and the AI returned personalized recommendations with restaurant cards.





AI Chat - Multi Turn

Followed up with a different query. The AI remembered the previous conversation context and switched recommendations.

New Chrome available

AI Assistant

AI Assistant

Clear Chat

Find dinner tonight | Best rated near me | Vegan options

What about something vegan and casual

Vegan and casual sounds like a great combo. Based on your preferences, I'd recommend the following options: 1. Green Leaf Cafe (4.0★, $) - This 100% vegan cafe in San Jose offers a variety of plant-based dishes in a casual atmosphere. They use fresh organic ingredients, and it's a great spot for a relaxed meal. 2. Veggie Delight (4.4★, $$) - While not entirely casual, Veggie Delight in San Jose has a creative plant-based menu with extensive vegan and vegetarian options. They have a cozy atmosphere, and it's a great place to try something new. Both of these options should fit the bill for a casual, vegan meal. If you're looking for something a bit more laid-back, Green Leaf Cafe might be the way to go. If you're open to a slightly more upscale atmosphere, Veggie Delight is a great choice. Keep in mind that these options might not be strictly Indian, as you initially mentioned, but they do offer delicious vegan cuisine. If you'd like to explore more options or stick to Indian cuisine, just let me know!

e.g. I want something romantic tonight...

Send

New Chrome available

AI Assistant

AI Assistant

Clear Chat

Find dinner tonight | Best rated near me | Vegan options

upscale atmosphere, Veggie Delight is a great choice. Keep in mind that these options might not be strictly Indian, as you initially mentioned, but they do offer delicious vegan cuisine. If you'd like to explore more options or stick to Indian cuisine, just let me know!

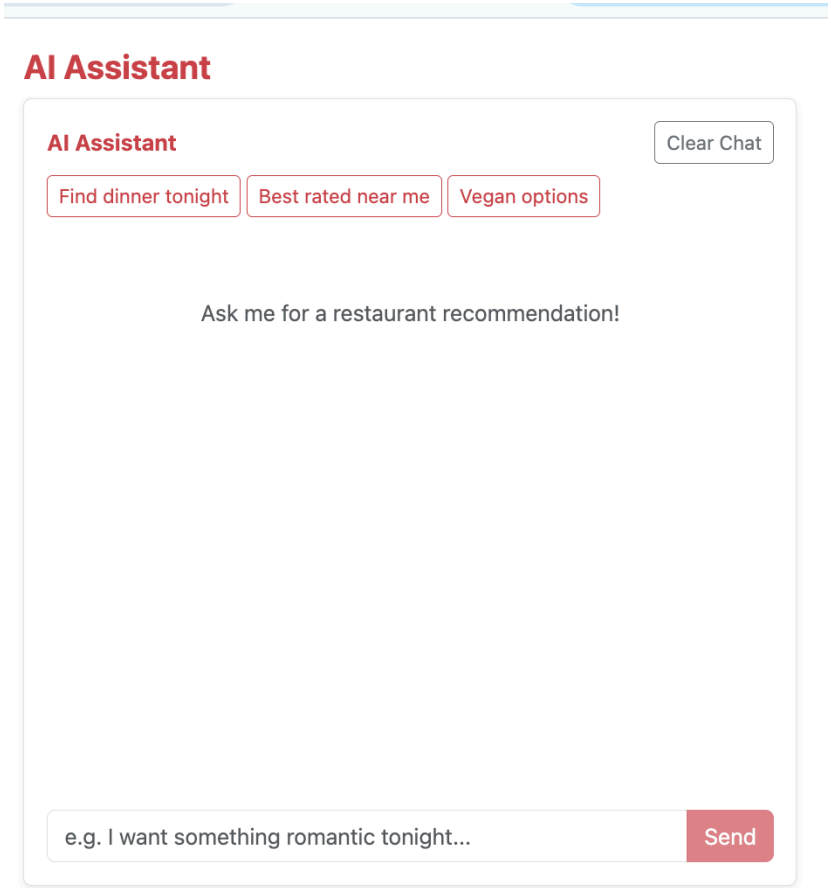
Green Leaf Cafe
American · $
4

Veggie Delight
American · $$
4.4

e.g. I want something romantic tonight...

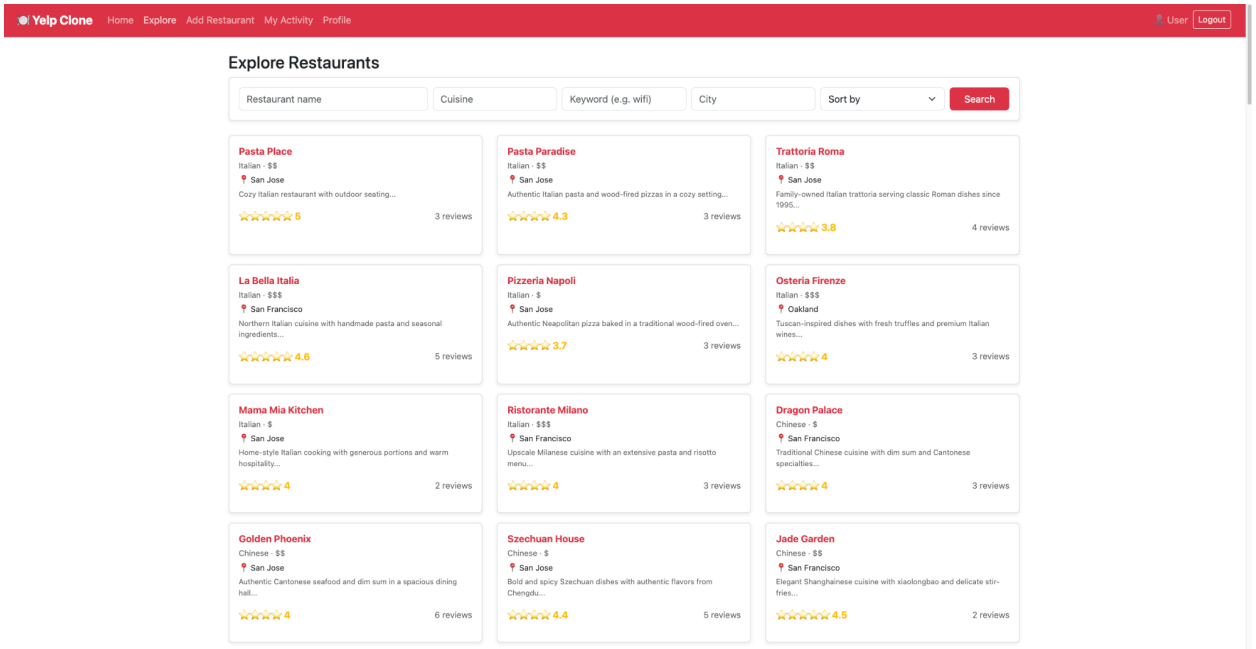
Send

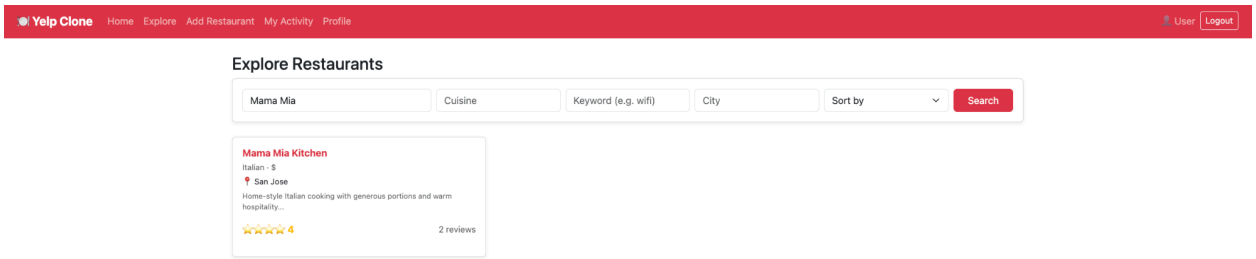
User can clear ai chat



Explore / Search Page

Search by name, cuisine, keyword and city. Sort dropdown auto-triggers without clicking Search.

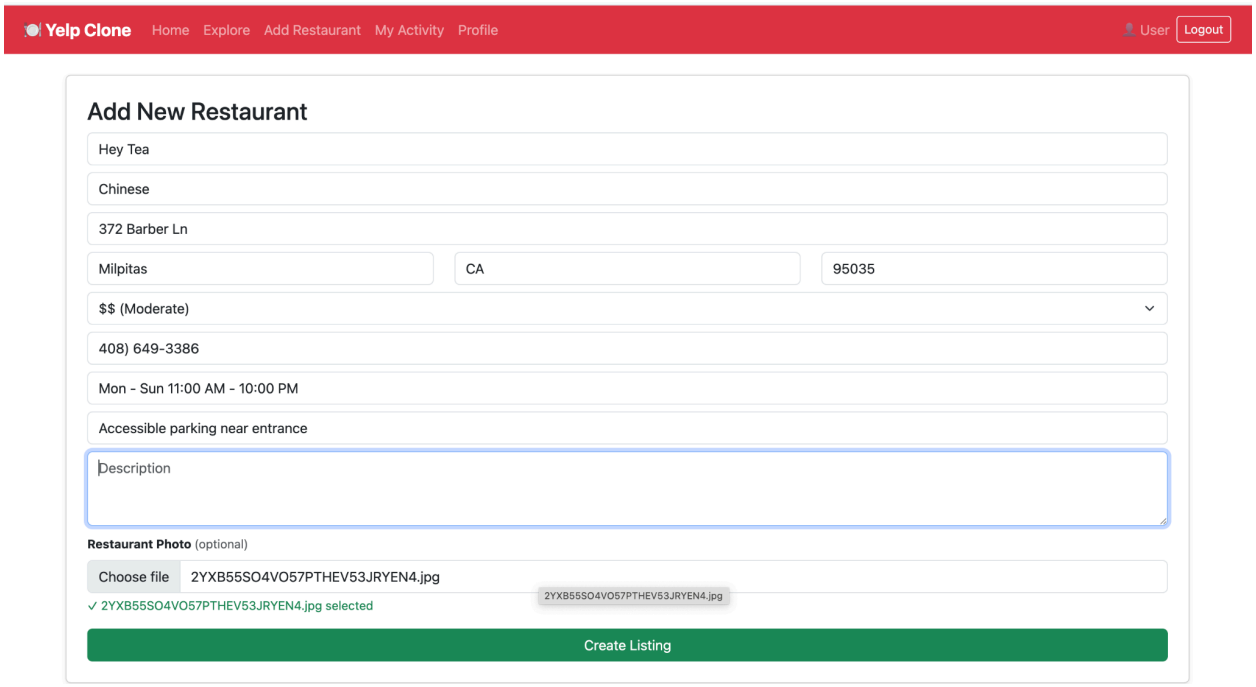




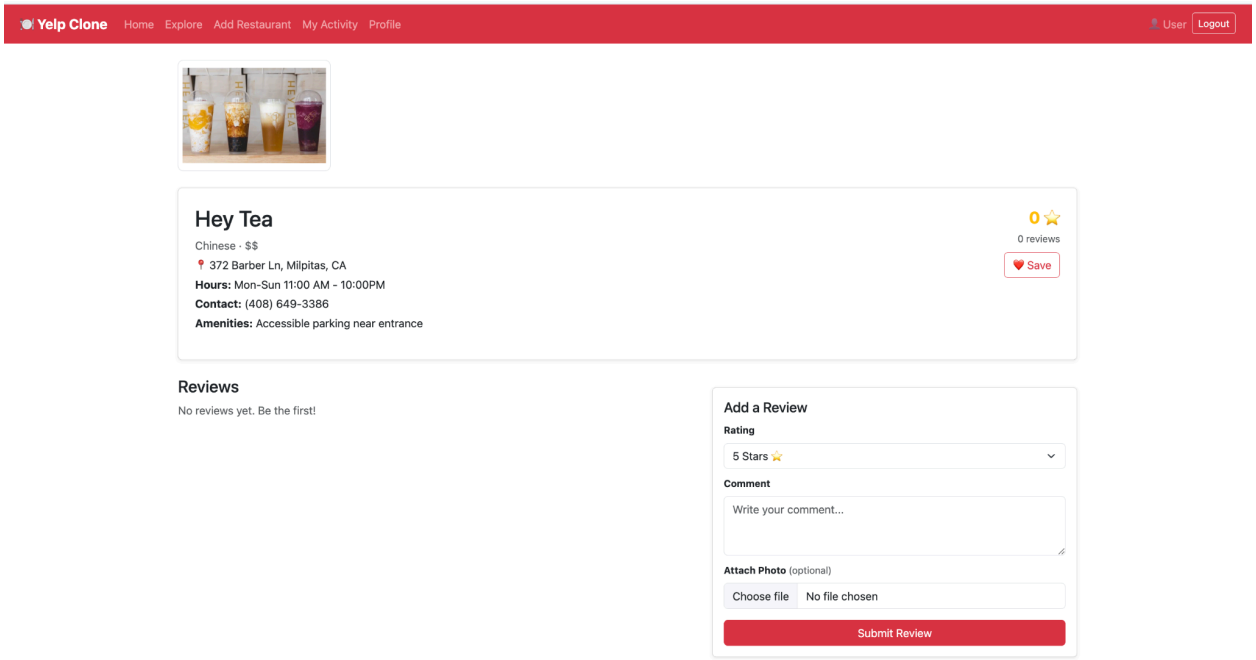
Restaurant Details Page

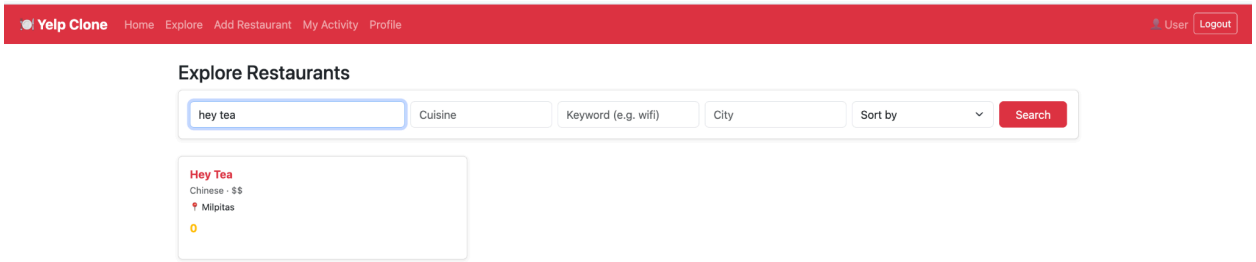
Shows full restaurant info, reviews with reviewer names and dates, and the write review form with optional photo upload.

Restaurant Form



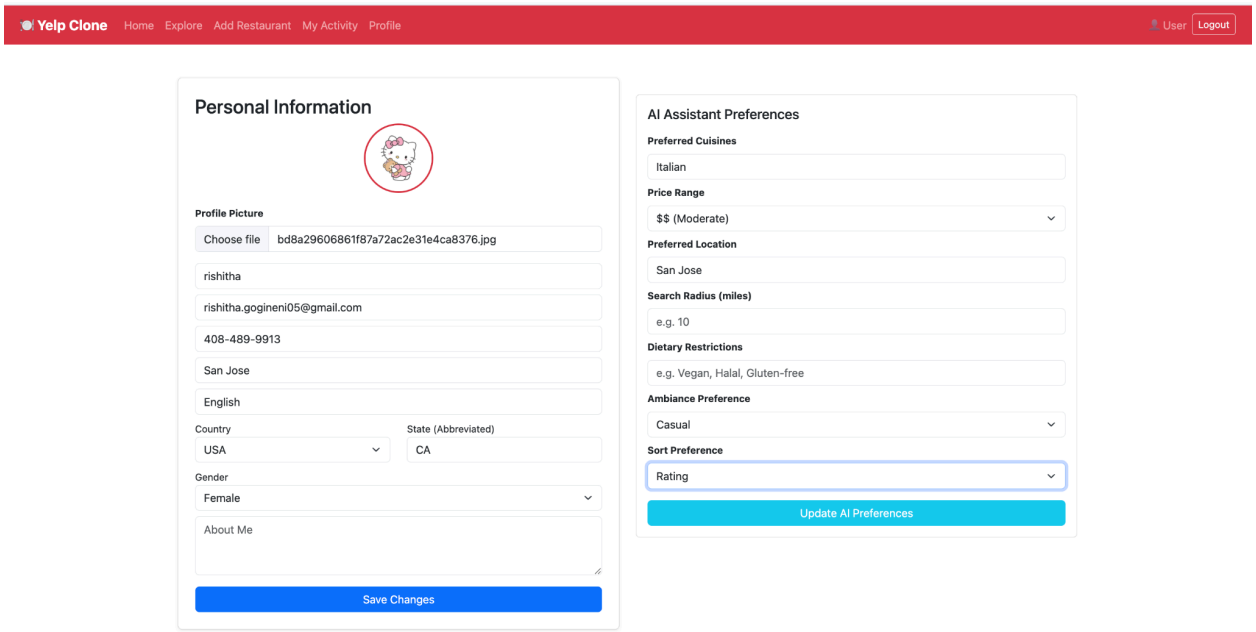
Added Restaurant successfully





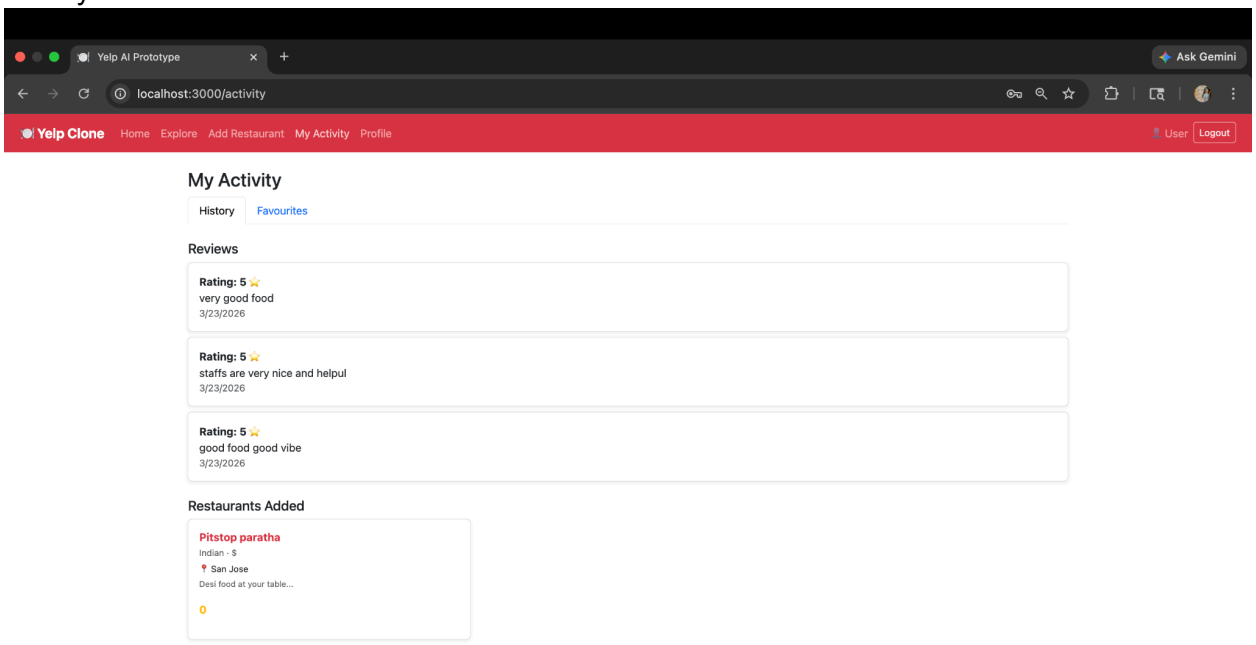
Profile and Preferences

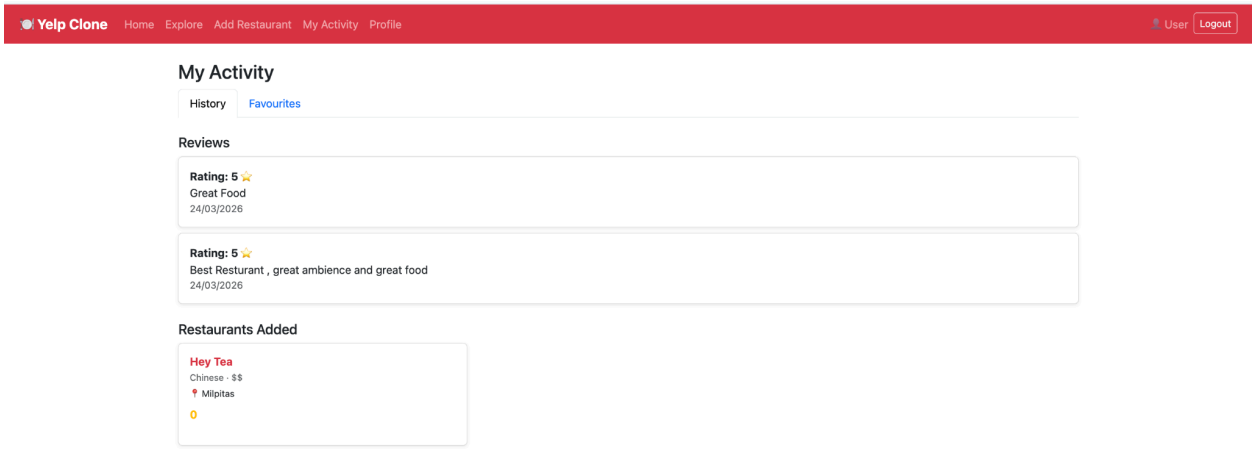
Country is a dropdown, state is validated to 2 letters. AI preferences are saved and used by the chatbot.

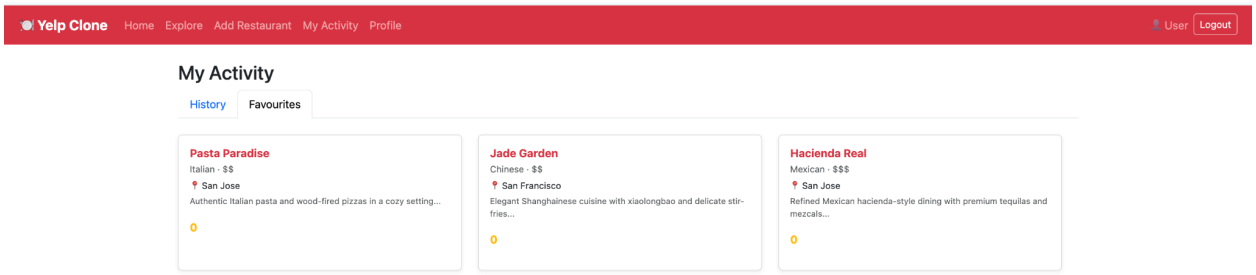


My Activity

History tab shows reviews and restaurants added. Favourites tab shows saved restaurants.

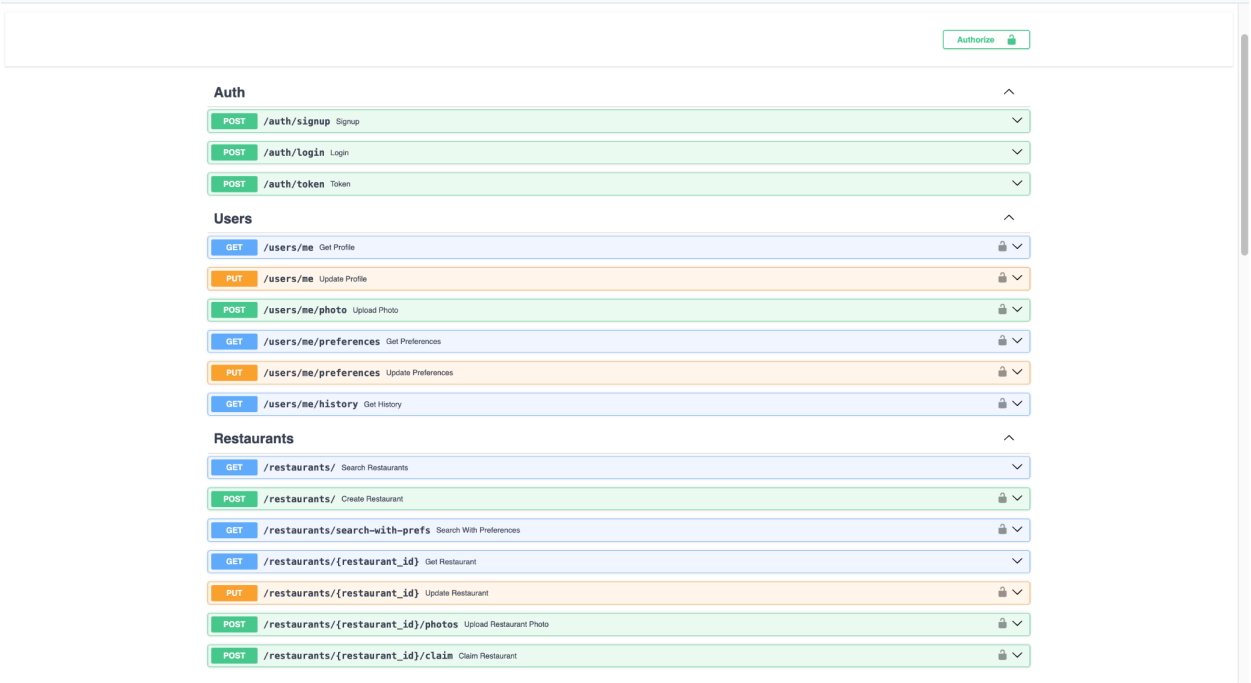






Swagger API Documentation

FastAPI auto-generates Swagger at http://localhost:8000/docs. All endpoints can be tested from the browser.



Reviews

- POST /reviews/ Create Review
- GET /reviews/restaurant/{restaurant_id} Get Reviews
- PUT /reviews/{review_id} Update Review
- DELETE /reviews/{review_id} Delete Review
- POST /reviews/{review_id}/photos Upload Review Photo

Favorites

- POST /favorites/{restaurant_id} Add Favorite
- DELETE /favorites/{restaurant_id} Remove Favorite
- GET /favorites/ Get Favorites

AI Assistant

- POST /ai-assistant/chat Chat
- GET /ai-assistant/history Get Chat History
- DELETE /ai-assistant/history Clear Chat History

Owner

- GET /owner/dashboard Owner Dashboard
- POST /owner/restaurants Post Restaurant
- PUT /owner/restaurants/{restaurant_id} Edit Restaurant
- POST /owner/restaurants/{restaurant_id}/photos Upload Owner Photo
- GET /owner/restaurants/{restaurant_id}/reviews Get Restaurant Reviews

default

- GET / Root

POST /auth/login - successful login returning JWT token

POST /auth/login Login

Parameters

Cancel Reset

No parameters

Request body required

application/json

Edit Value | Schema

```
{
  "email": "rishitha.gogineni@sjsu.edu",
  "password": "Rishitha123"
}
```

Execute Clear

Responses

Server response

Code	Details
200	Response body


```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiIxMiIsImV4cCI6MTc3NDMyMDA1Nn0._TTBSYNHn71DJwztp_dbyQXMDrqJHHy9NsnZL013aRE",
  "token_type": "bearer",
  "role": "user"
}
```


Download

Response headers

```
access-control-allow-credentials: true
access-control-allow-origin: http://localhost:8000
content-length: 175
content-type: application/json
date: Tue,24 Mar 2026 01:40:55 GMT
server: uvicorn
vary: Origin
```

Responses

GET /restaurants/ - search returning restaurant list

200

Response body


```
[
  {
    "restaurant": {
      "name": "Pasta Place",
      "contact": "408-123-4567",
      "cuisine_type": "Italian",
      "pricing_tier": "$$",
      "address": "123 Main St",
      "amenities": "wifi, outdoor seating",
      "city": "San Jose",
      "is_claimed": false,
      "state": "CA",
      "created_at": "2026-03-19T16:24:56",
      "zip": "95101",
      "owner_id": 1,
      "description": "Cozy Italian restaurant with outdoor seating",
      "id": 1,
      "hours": "Mon-Sun 11am-10pm"
    },
    "avg_rating": 5,
    "review_count": 3
  },
  {
    "restaurant": {
```


Download

Response headers

```
content-length: 44271
content-type: application/json
date: Tue,24 Mar 2026 01:42:42 GMT
server: uvicorn
```

POST /reviews/ - creating a review successfully

POST /reviews/ Create Review

Parameters

Cancel Reset

No parameters

Request body required

application/json

Edit Value | Schema


```
{
  "restaurant_id": 1,
  "rating": 5,
  "comment": "Good ambience"
}
```

Curl

```
curl -X 'POST' \
  'http://localhost:8000/reviews/' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiIxMiIsImV4cCI6MTc3NDMyMDI5Nn0.kCFz5u1G9z1x5nY1iVk8F9eZeo6h3YS3lBYDS4JgaTY' \
  -H 'Content-Type: application/json' \
  -d '{
  "restaurant_id": 1,
  "rating": 5,
  "comment": "Good ambience"
}'
```

Request URL

```
http://localhost:8000/reviews/
```

Server response

Code	Details
200	Response body

```
{
  "id": 331,
  "rating": 5,
  "review_date": "2026-03-24T01:45:04",
  "user_id": 12,
  "restaurant_id": 1,
  "comment": "Good ambience",
  "updated_at": "2026-03-24T01:45:04"
}
```

Download

Response headers

```
access-control-allow-credentials: true
access-control-allow-origin: http://localhost:8000
content-length: 149
content-type: application/json
date: Tue,24 Mar 2026 01:45:03 GMT
server: uvicorn
vary: Origin
```

Responses

POST /ai-assistant/chat - AI chat response

AI Assistant

POST /ai-assistant/chat Chat

Parameters

Cancel Reset

No parameters

Request body required

application/json

Edit Value | Schema

```
{
  "message": "I want to try mexican",
  "conversation_history": []
}
```

Execute Clear

Responses

Code	Details
200	Response body


```
{
  "response": "Mexican food is delicious. However, I noticed you initially mentioned you're in the mood for Italian cuisine, but since you'd like to try Mexican now, I'd be ha
ppy to help.\n\nConsidering your preferences for a casual ambiance and a $$ price tier, I found a few Mexican options in the San Jose area. Unfortunately, none of the availabl
e options are vegan, but I can suggest some popular ones. \n\nYou might enjoy Taqueria El Sol (4.2★, $) in San Jose, it's a neighborhood taqueria with freshly made tortillas a
nd traditional fillings. Another option could be El Rancho Grill (3.7★, $) in San Jose, which offers ranch-style Mexican BBQ with slow-cooked meats and fresh salsas.\n\nIf yo
u're open to exploring nearby cities, The Taco Stand (4.2★, $) in Oakland is a great option for simple and delicious street-style tacos.\n\nKeep in mind that these options may
not exactly fit your initial preferences, but they're highly rated and popular among locals. Would you like me to look for more options or provide more information about these
suggestions?",
  "recommendations": [
    {
      "id": 22,
      "name": "El Rancho Grill",
      "cuisine": "Mexican",
      "city": "San Jose",
      "pricing_tier": "$",
      "description": "Ranch-style Mexican BBQ with slow-cooked meats and fresh salsas",
      "avg_rating": 3.7
    },
    {
      "id": 58,
      "name": "Taqueria El Sol",
      "cuisine": "Mexican",
      "city": "San Jose",
      "pricing_tier": "$",
```


Download

Response headers

```
access-control-allow-credentials: true
access-control-allow-origin: http://localhost:8000
content-length: 2320
content-type: application/json
date: Tue,24 Mar 2026 01:46:54 GMT
server: uvicorn
vary: Origin
```

Responses

200	Response body


```
    {
      "id": 58,
      "name": "Taqueria El Sol",
      "cuisine": "Mexican",
      "city": "San Jose",
      "pricing_tier": "$",
      "description": "Neighborhood taqueria with freshly made tortillas and traditional fillings",
      "avg_rating": 3.8
    },
    {
      "id": 89,
      "name": "The Taco Stand",
      "cuisine": "Mexican",
      "city": "Oakland",
      "pricing_tier": "$",
      "description": "Simple and delicious street-style tacos with homemade salsas and fresh tortillas",
      "avg_rating": 4.2
    }
  ],
  "web_context": "Guide to Mexican food\nGuide to Mexican food\n\nA burrito will often be enough for a whole meal.\n\nInterested in taking a Food tour while in Mexico City? Sh
erpa Food Tours are one of the best options, you'll try out 10+ dishes while also exploring some of the best spots in the Roma Norte neighbourhood (Mexico City's foodie destin
ation). More here.\n\n## Fajitas\n\nFajitas are served with the same large flour tortilla as burritos. The difference is the filling and how it is served. [...] Frank (bbqboy)
\n\nAugust 30, 2023 at 12:14 pm\n\nThank you for that. I see it originated in Patzcuaro - where we'll be s"
}
```


Download

Response headers

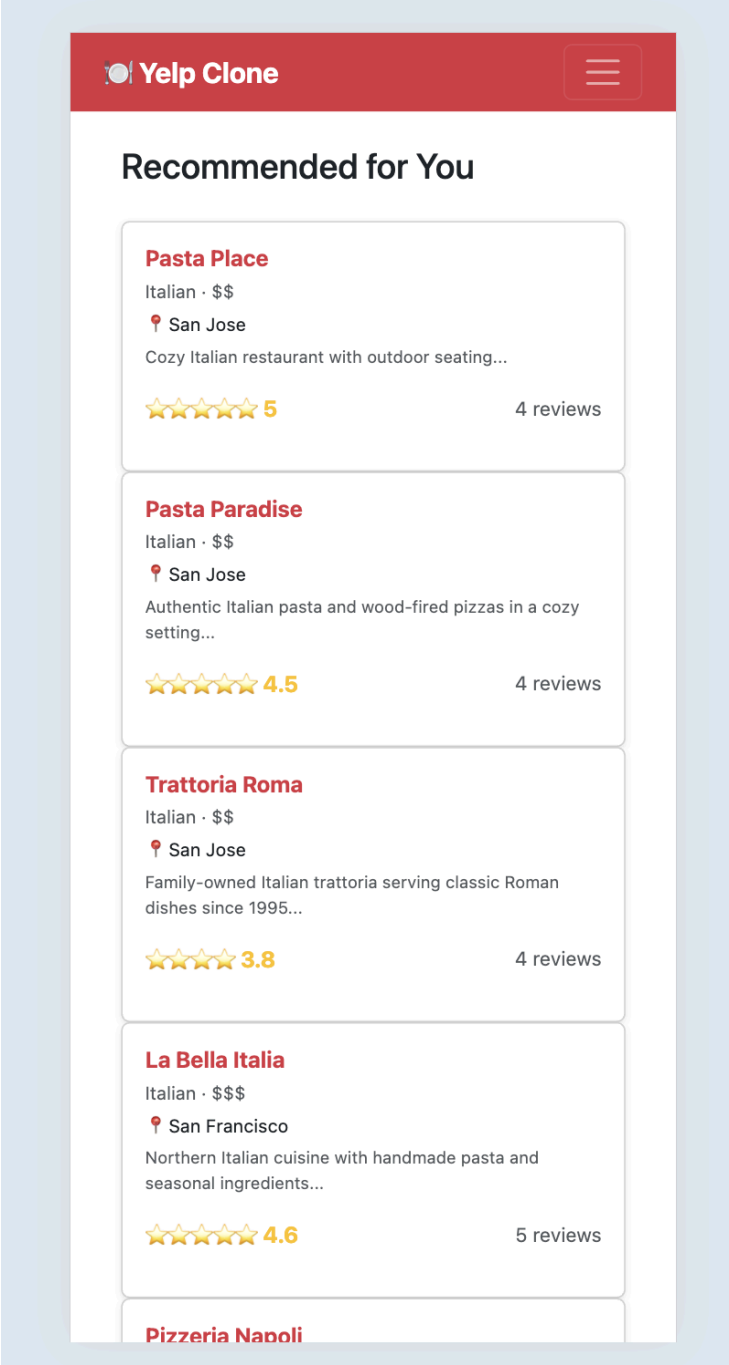
```
access-control-allow-credentials: true
access-control-allow-origin: http://localhost:8000
content-length: 2320
content-type: application/json
date: Tue,24 Mar 2026 01:46:54 GMT
server: uvicorn
vary: Origin
```

Responses

Code	Description	Links

Mobile View

Navbar collapses to hamburger menu on mobile. Restaurant cards stack in a single column. Bootstrap handles the responsive layout.



Conclusion

The Yelp prototype was completed with all the required features. The backend has full REST APIs for auth, profiles, restaurants, reviews, favourites, owner features and the AI assistant. The frontend connects to all of them and the whole thing works end to end.

The AI assistant was the most challenging part. Using LangChain with Groq to do proper NLU-based filter extraction instead of keyword matching made a real difference in how well it

handles different types of queries. The multi-turn conversation also works well in practice as we tested during development.

References

- FastAPI - https://fastapi.tiangolo.com
- LangChain - https://python.langchain.com
- Groq API - https://console.groq.com
- Tavily Search - https://www.tavily.com
- React - https://react.dev
- SQLAlchemy - https://docs.sqlalchemy.org
- Bootstrap - https://getbootstrap.com
- Yelp reference - https://www.yelp.com